App Name: Smart Pantry Manager

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Module Name: Mobile App Development

GITHUB LINK: https://github.com/wandile2003/Smart-Pantry-Manager

VIDEO LINK: https://drive.google.com/file/d/1Ddip30N8JnDptUbP_mtnsh_E4VPp7H5M/view?usp=drivesdk

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Introduction

- The problem here is the Food Waste since a person can`t choose what to cook or make .
- Smart Pantry Manager is a Java Android application intended to help a household use ingredients already available at home and reduce avoidable food waste. A user can record pantry ingredients with quantities, units and optional expiry dates. The application suggests a recipe only when all its required ingredients are present in sufficient quantities. A recipe with even one missing or insufficient ingredient must not appear in the strict suggestions list.

Objectives

- Offline pantry and recipe Suggestions(Can make Now /Missing 1)

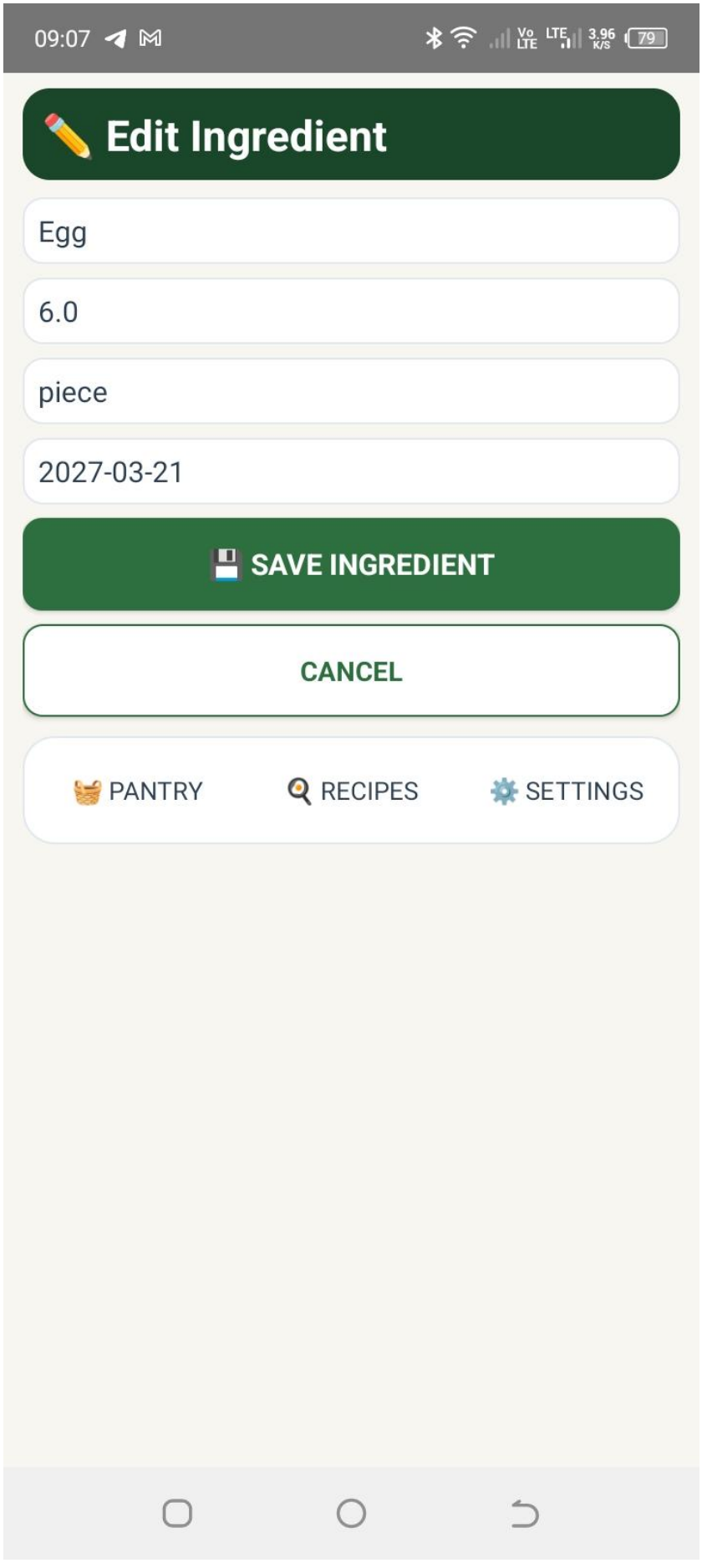
- Scope : Java, SQLite, Android 7+

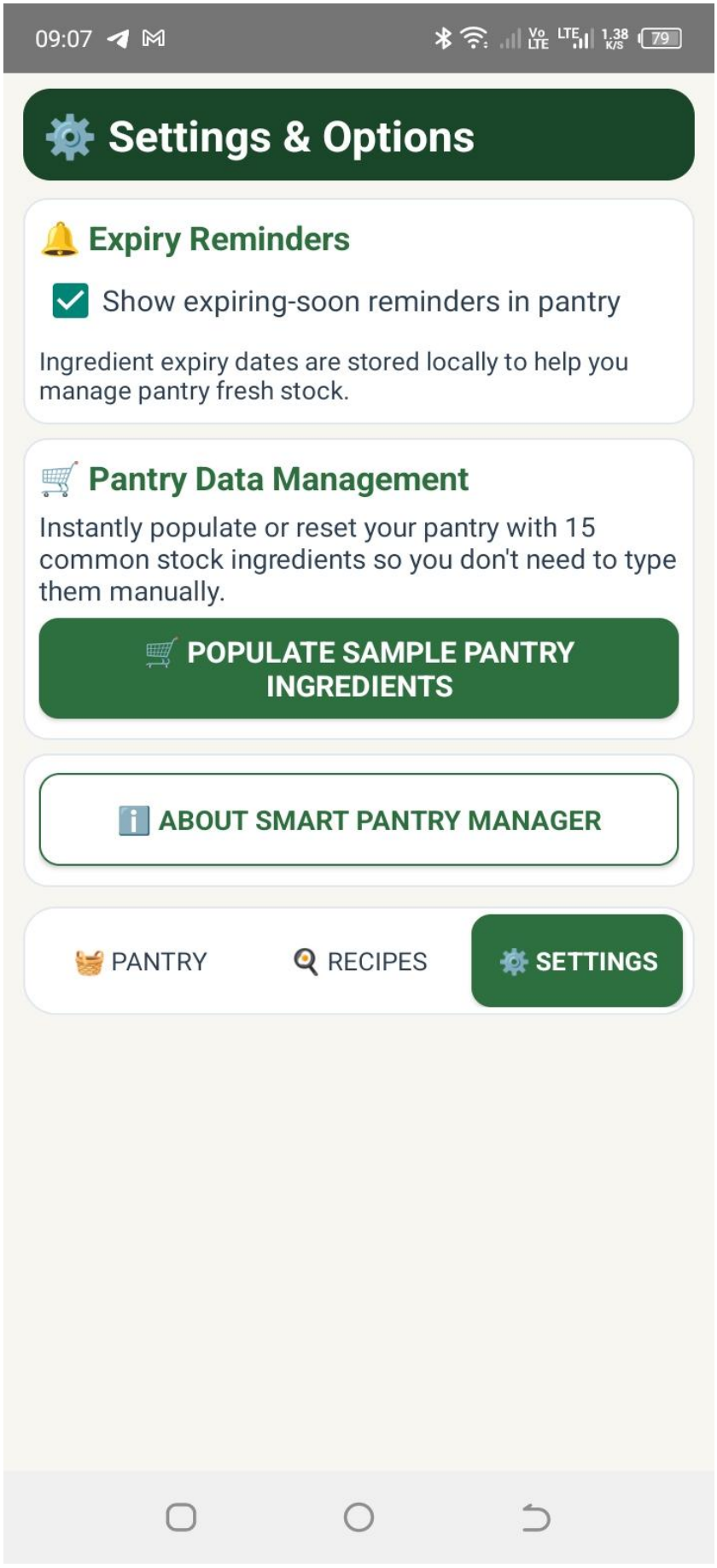
Requirements

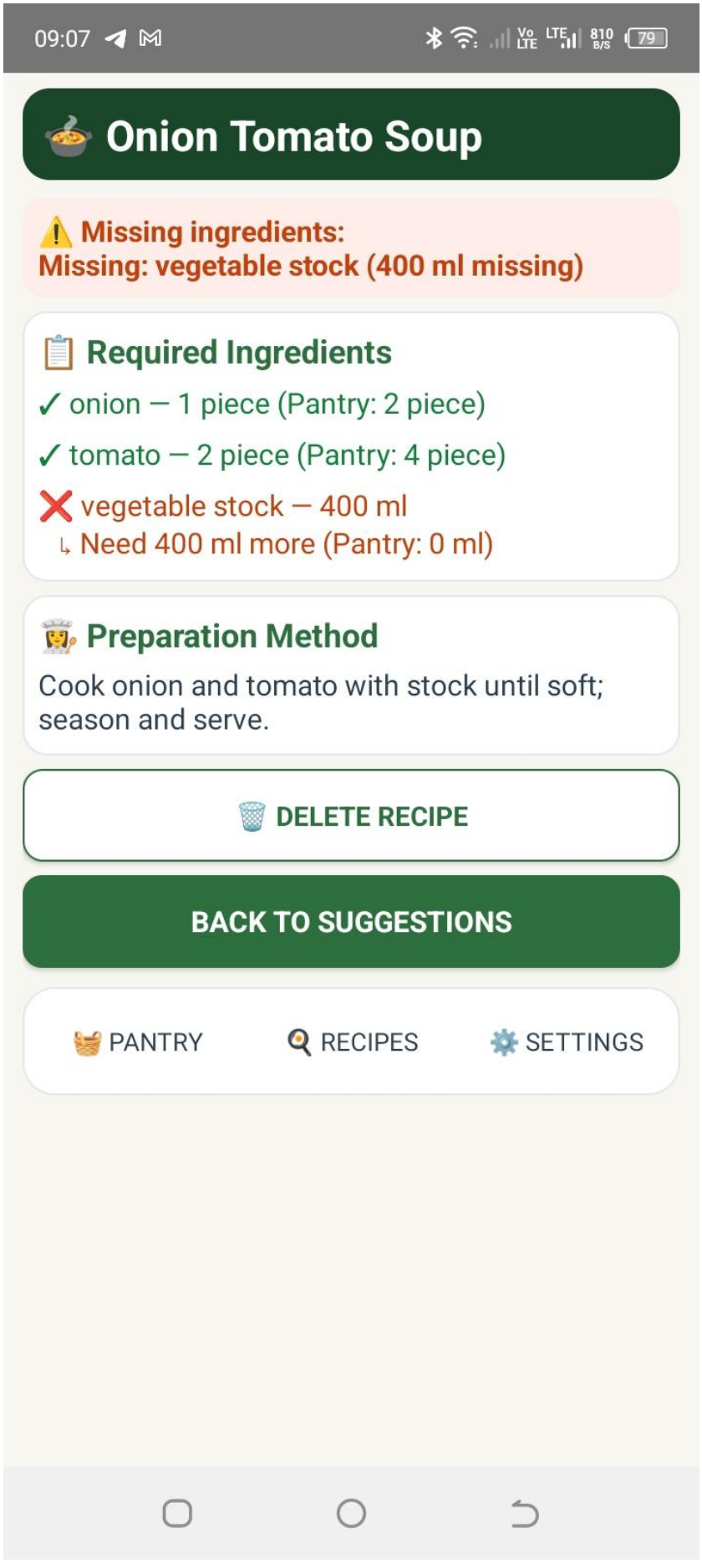
- Table with FR 1 -FR 5 (Manage Pantry, View Pantry, Suggest Strict, Suggest Almost, Navigation).
- Non- Functional: offline, Material Design,<1s Matching.

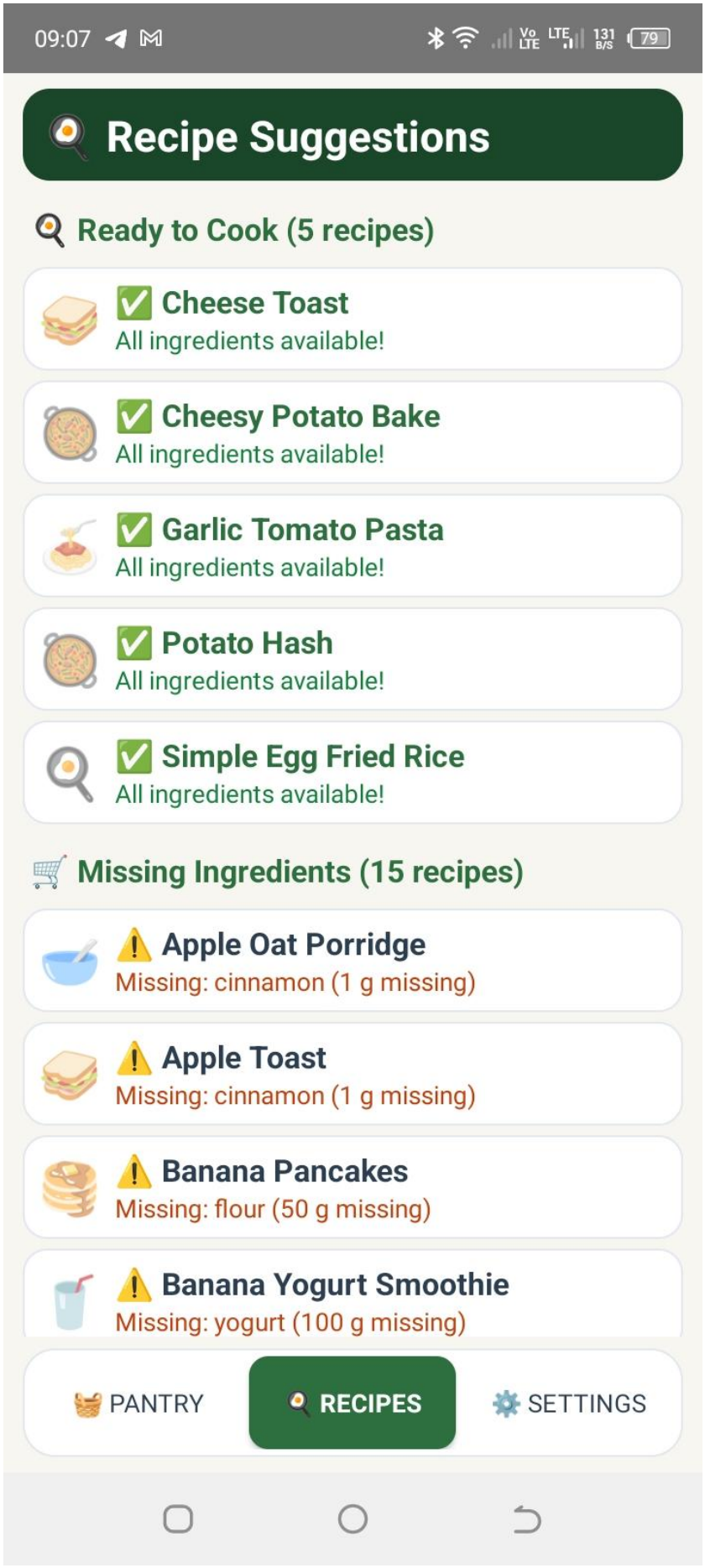
System Design

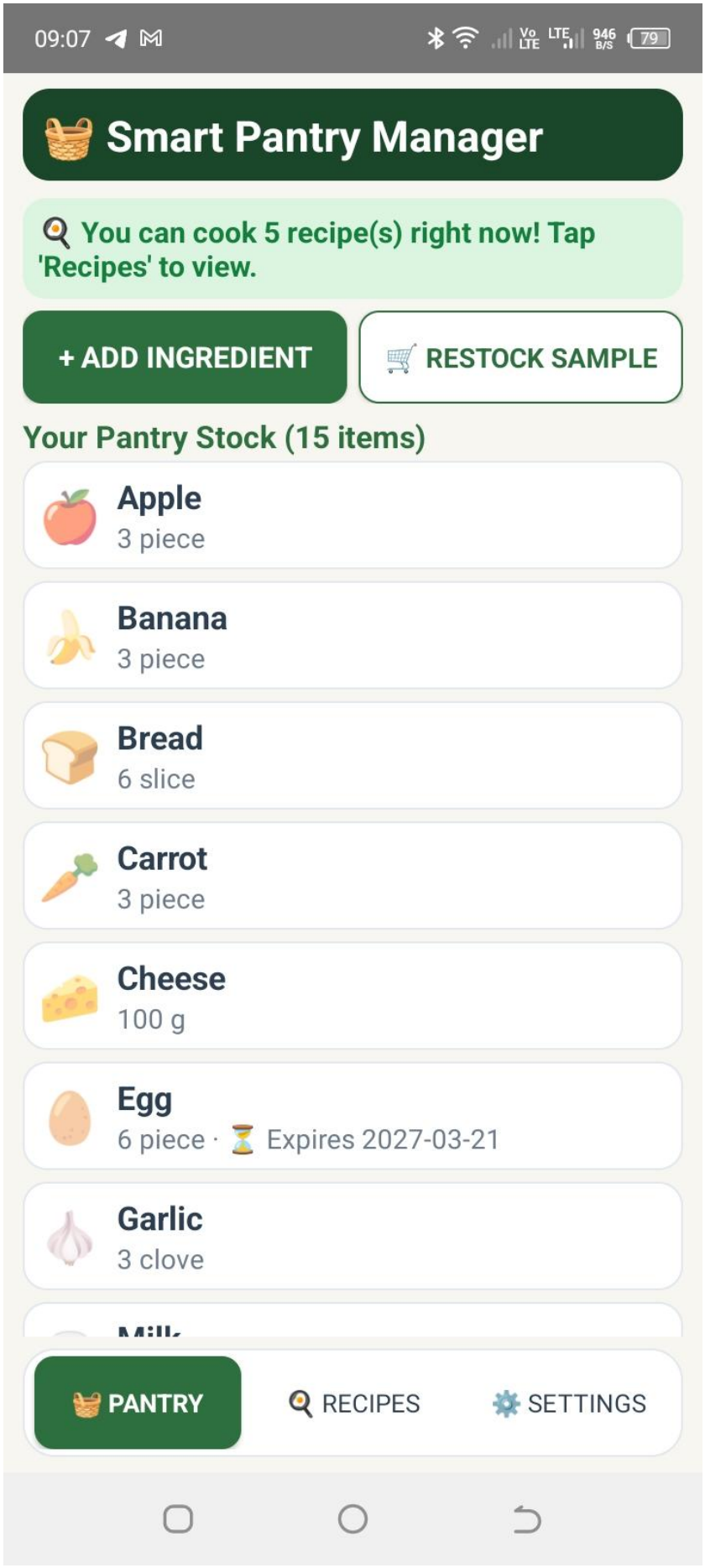
- 3.1 Architecture: Single Activity+ 3 fragments+ BottomNavigationView
- 3.2 Database Design: 3 Table- pantry_items, recipes, recipe_ingredients(with ER diagram placeholder)
- UI design: 4 Screenshot
- Pantry List → Add/Edit Ingredient → Pantry List. Pantry List → Suggested Recipes → Recipe Detail. Pantry List, Suggested Recipes and Recipe Detail also provide navigation to Settings. (Developers, 2026)


09:07
Edit Ingredient
Egg
6.0
piece
2027-03-21
SAVE INGREDIENT
CANCEL
PANTRY
RECIPES
SETTINGS


09:07
Settings & Options
Expiry Reminders
Show expiring-soon reminders in pantry
Ingredient expiry dates are stored locally to help you manage pantry fresh stock.
Pantry Data Management
Instantly populate or reset your pantry with 15 common stock ingredients so you don't need to type them manually.
POPULATE SAMPLE PANTRY INGREDIENTS
ABOUT SMART PANTRY MANAGER
PANTRY
RECIPES
SETTINGS


09:07
Onion Tomato Soup
Missing ingredients:
Missing: vegetable stock (400 ml missing)
Required Ingredients
✓ onion — 1 piece (Pantry: 2 piece)
✓ tomato — 2 piece (Pantry: 4 piece)
❌ vegetable stock — 400 ml
↳ Need 400 ml more (Pantry: 0 ml)
Preparation Method
Cook onion and tomato with stock until soft; season and serve.
DELETE RECIPE
BACK TO SUGGESTIONS
PANTRY
RECIPES
SETTINGS


09:07
Recipe Suggestions
Ready to Cook (5 recipes)
✅ Cheese Toast
All ingredients available!
✅ Cheesy Potato Bake
All ingredients available!
✅ Garlic Tomato Pasta
All ingredients available!
✅ Potato Hash
All ingredients available!
✅ Simple Egg Fried Rice
All ingredients available!
Missing Ingredients (15 recipes)
⚠️ Apple Oat Porridge
Missing: cinnamon (1 g missing)
⚠️ Apple Toast
Missing: cinnamon (1 g missing)
⚠️ Banana Pancakes
Missing: flour (50 g missing)
⚠️ Banana Yogurt Smoothie
Missing: yogurt (100 g missing)
PANTRY
RECIPES
SETTINGS


09:07
Smart Pantry Manager
You can cook 5 recipe(s) right now! Tap 'Recipes' to view.
+ ADD INGREDIENT
RESTOCK SAMPLE
Your Pantry Stock (15 items)
Apple
3 piece
Banana
3 piece
Bread
6 slice
Carrot
3 piece
Cheese
100 g
Egg
6 piece · Expires 2027-03-21
Garlic
3 clove
PANTRY
RECIPES
SETTINGS

2.2 Database model

The local SQLite database contains three tables:

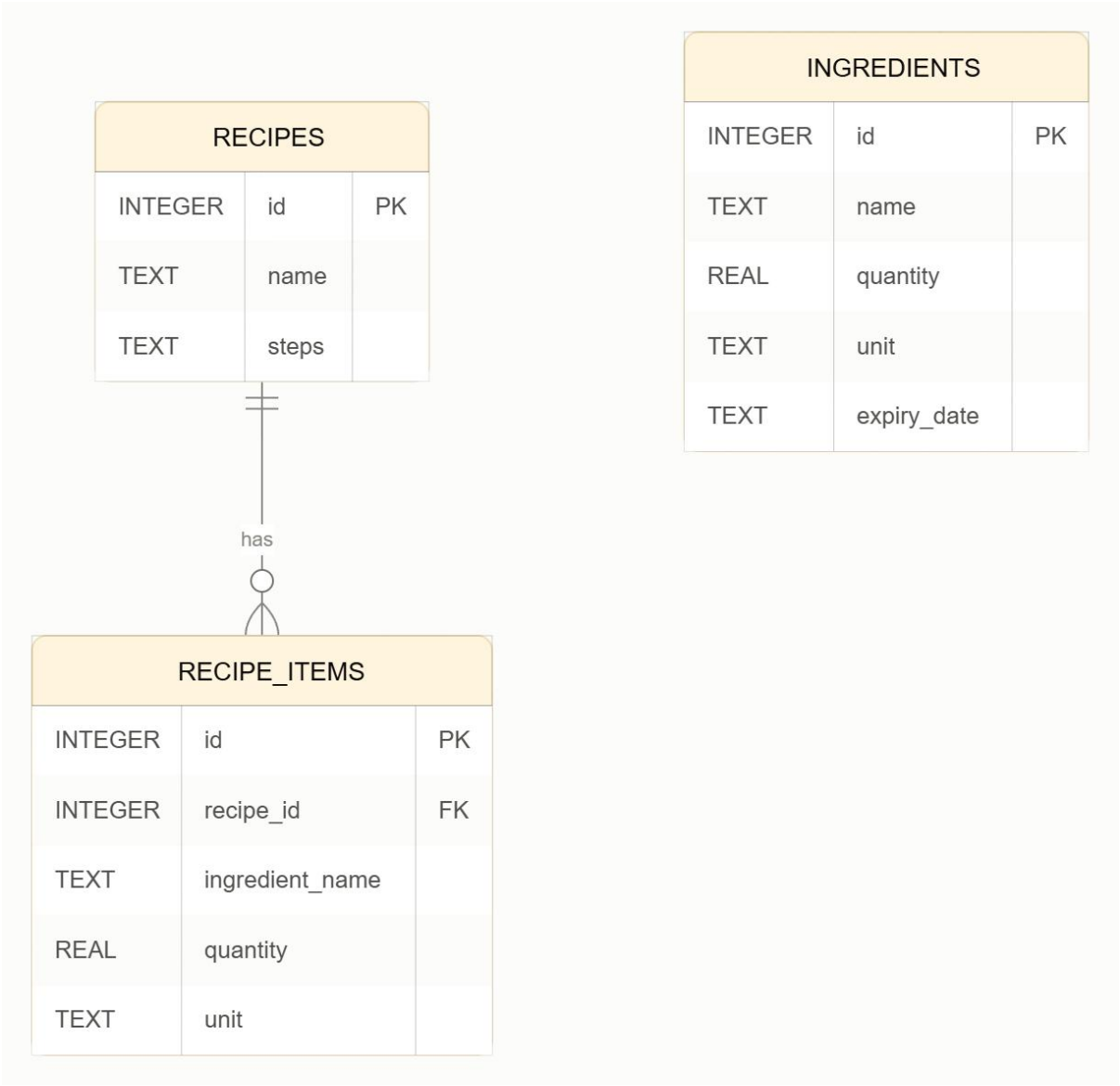
- `ingredients(id, name, quantity, unit, expiry_date)` — user-managed pantry records.

- `recipes(id, name, steps)` — preloaded recipe name and method.

- `recipe_items(id, recipe_id, ingredient_name, quantity, unit)` — required ingredients linked to a recipe by `recipe_id`.

`recipes` has a one-to-many relationship with `recipe_items`. Pantry data is independent and is compared with recipe requirements in Java. SQLite was selected because the assignment allows local persistence, the app is focused on one device, and it avoids needing a remote server or account.

ER DIAGRAM



Implementation

- Tools list+ Key Files explanation(MainActivity, Datahelper, PantryFragment, SuggestedFragment, Bottom_menu_xml)
- Space for your actual code snippet from SuggestedFragment

Testing

- Ready-made table with 5 test cases using the exact data which I used earlier
- TC 2: Add tomato 2, egg 3, onion 1= Tomato Omelette in can make now.
- TC 3: Delete onion= moves to Almost missing onion
- pantry list, add form, validation error, created item, edited item, deletion, recipe suggestion that qualifies, a recipe excluded when one ingredient is missing, recipe details, settings, and persistence after relaunch. (Developers, 2026) (Developers, Intent and Intent filters, 2026)

Challenges And Solutions

- The challenges I have been facing for the real build failed issues
- Package name mismatch fix
- Bottom_menu_xml missing fix
- FirstFragment/SecondFragment template bug
- Empty SQLite fix

The Strict Matching

The `getStrictMatches()` method reads pantry records and recipe requirements. It normalises simple singular/plural forms and common unit spellings. For each required ingredient it sums quantities whose normalised name matches and whose unit can be

converted safely. If the available amount is less than the required amount, or units are incompatible, the recipe is rejected. Only recipes for which every requirement passes are returned.

Conclusion

Smart Pantry Manager demonstrates Java Android activities, Intent-based navigation, dynamic list display, local SQLite persistence and a strict recipe-matching rule. The seeded recipe collection provides a starting point that can be expanded. Future improvements could include automated expiry notifications, richer unit conversion, and tests for more ingredient-name variations. Update this reflection to describe what you personally learned and what you verified during your demonstration.

Referencing

References

Developers, A. (2026, October 26). *Intent and Intent filters*. Retrieved from Intent and Intent filters: http://developer.android.com

Developers, A. (2026, October 26). *SQLite*. Retrieved from SQLite: http://developer.android.com